

UIDAI Data Hackathon 2026

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Project Title : Aadhaar Update Load Balancer (AULB)

Problem Statement

The services of Aadhaar enrolment and update are important in facilitating access of citizens to welfare schemes, banking, education and digital services. Nevertheless, Aadhaar centres are regularly characterized by long queues, workload imbalances, missed deadlines on update, and bottlenecks in services. The problems with these are that enrolment and update infrastructure is operated in a very reactive way, i.e. resources are only changed once congestion has been experienced.

The specified problem statement presupposes the identification of meaningful trends, anomalies, and predictive indicators in Aadhaar enrolment and update data and converting them into practical insights that can be used to make informed decisions and improve the system.

Proposed Approach

In this project, the proposed solution is **AULB (Aadhaar Update Load Balancer)**, which is a predictive analytics and decision-support system that converts the past data of Aadhaar enrolment, demographic update, and biometric update into future operational intelligence. The approach does not just end at the descriptive analysis but completes the data → prediction → decision loop.

The system predicts the pressure of Aadhaar update at the PIN-code level, identifies the abnormal patterns of services, and suggests proactive administrative measures, including mobile enrolment unit deployment, increasing the working hours of centres, or redistributing operators. This will allow UIDAI to shift to reactive service management to predictive and optimized planning.

Dataset Description:

1) Biometric

date	state	district	pincode	bio_age_5_	bio_age_17_
19-09-2025	Goa	North Goa	403502	0	4
19-09-2025	Goa	North Goa	403508	1	4
19-09-2025	Goa	North Goa	403513	2	0
19-09-2025	Goa	North Goa	403527	2	2
19-09-2025	Goa	South Goa	403601	7	3

This data contains biometric data at the pincode level on a given date. Every row is a distinct area in which pincode is famous, the state, and the district of that area. The data will consist of population distributions by age (or demographic totals) of various age groups (e.g. the values below bioage5 and bioage17, which will be the number of persons who fall into the relevant age groups in that region). On the whole, it assists in analyzing and comparing demographic distribution in terms of age among various pincodes in districts such as states over time.

2)Demographic

date	state	district	pincode	bio_age_5_	bio_age_17_
19-09-2025	Goa	North Goa	403502	0	4
19-09-2025	Goa	North Goa	403508	1	4
19-09-2025	Goa	North Goa	403513	2	0
19-09-2025	Goa	North Goa	403527	2	2
19-09-2025	Goa	South Goa	403601	7	3

This data set contains demographic data that has been registered around various regions on a particular date. Different rows are associated with a given area that is represented by pincode and state as well as district. The columns demoage and demoage17_ are displaying the demographic counts per age in that place or the number of persons aged in the respective age groups. On the whole, this data set assists in analyzing and comparing between the population distribution between several districts and states according to the age groups of the recorded date.

3)Enrollment dataset

This enrollment record is a set of region-specific enrollment records of particular dates. The rows constitute enrollment data on a given pincode with the state and district of the pincode. The data is divided into various age groups using the age05, age517, and age18greater columns, which represent how many people have been enrolled under the age groups age 05-5 years, age 5-17 years and age 18 and above respectively. In general, it assists in the analysis and comparison of the enrollment distribution at various locations and age groupings in the course of time.

Methodology:

The methodology used in this project is a multi-stage analytical pipeline that is structured in a way that it can extract meaningful patterns in Aadhaar data and transforming these patterns into predictive and actionable insights. The model is a combination of preprocessing of data, features engineering, statistical data analysis, time series prediction, and optimization of decisions based on rules.

Data Cleaning and Preprocessing:

Raw Aadhaar data were initially cleaned and made to be standardized to be accurate, consistent and reproducible. Major preprocessing procedures were:

- Elimination of duplicated records that occur because of multiple aggregation records.
- Normalization of names of states and districts to eliminate irregularities in the spelling and forms.
- Checking and screening of PIN codes to only keep valid geographic identifiers.
- Date fields converted to a time-series format to make them amenable to time analysis.
- Dealing with missing or zero values with logical defaults based on availability flags of updates.

The Aadhaar enrolment, demographic update and biometric update datasets were then combined into a single Aadhaar master dataset using common keys (date, state, district, pincode). This composite

dataset was able to conduct a cross-domain analysis of enrolment, demographic change and biometric activity.

Exploratory Data Analysis (EDA):

Exploratory analysis was used to learn about baseline trends and distributions:

1. Univariate analysis was used to investigate volumes of enrolment and update by age, region and time.
2. The bivariate analysis examined the correlations between enrolment volume and demographic or biometric updates.
3. Trivariate analysis was used to determine the combined effect of age, geography, and update type on service demand. This discussion demonstrated a high level of temporal and spatial variability, which drove the desire to model the predictability instead of reporting it.

Feature Engineering and Load Modeling (Phase 1)

To get out of the number of raw numbers the project constructed domain metrics of Aadhaar service pressure:

- Age-Transition Pressure Index (ATPI): Measures the demand of biometric update due to the migration of residents to mandatory biometric age groups.
- Demographic Volatility Index (DVI): The degree of demographic updates including those of migration and changes caused by life events.
- Biometric Stress Index (BSI): This is the weight of biometric refresh, correction and re-capture work.
- Population Pressure Metric: Latent future risk is captured with the use of log-scaled enrolment volume, in order not to dominate other indicators.

All the indicators were normalized to make them comparable and aggregated with weights to generate a decipherable Aadhaar Load

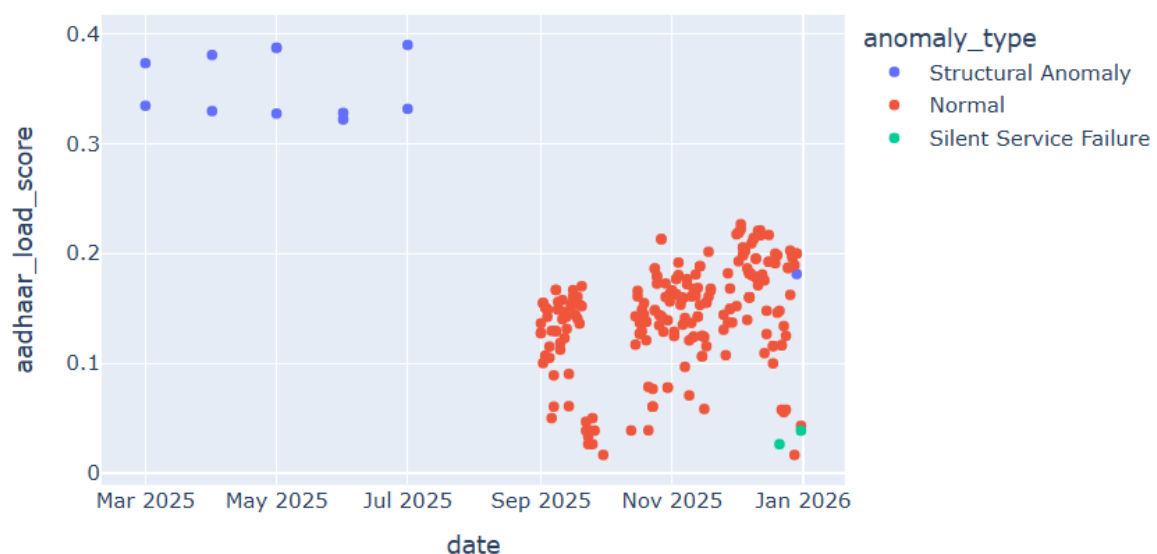
Score. This score allowed categorizing PIN codes as normal, emerging stress, and critical overload.

Anomaly Detection Framework (Phase 2)

Mean trends tend to conceal serious Aadhaar service interruptions. A hybrid anomaly detection framework was adopted to reveal the abnormal behaviour that could not be detected using aggregated statistics.

Anomaly Timeline visualization charts Aadhaar Load Score against time and points shown by the colour of anomaly type. Most observations are concentrated on normal functioning, and the exceptions are manifested as distinctly differentiated patterns.

Anomaly Timeline



Rolling means and z-scores can detect sudden overload spikes which represent short-term bursts in which the update demand exponentially exceeds historical levels. These come out as high points in the timeline, which means that there are times of stress in service that may lead to long queues and delays.

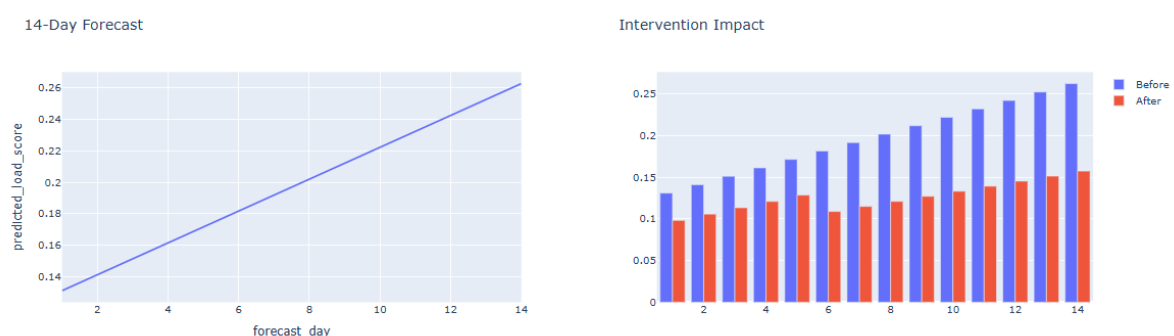
One of the innovations is **the identification of silent service failures**, in which the activity of updating Aadhaar plummets uncharacteristically low even when there is high population pressure. These arguments are found towards the bottom of the plot and tend to point at centre closures, system outages or manpower shortages as opposed to a lack of demand.

A multivariate Isolation Forest model is used to identify the complex and non-linear abnormal patterns by using load level, load change, and pressure indicators. This determines structural anomalies-isolated behaviours that are not in the normal combinations of operations even in situations where the raw load values are not extreme.

Phase 2, with the combination of statistical deviation, silent failure logic, and detection based on machine-learning, detects both visible overload events and hidden service disruptions, which serve as important inputs in subsequent phases to forecast and do operational planning.

Demand Forecasting and Early-Warning System (Phase 3)

Time-series forecasting models were used at the PIN-code level to forecast the Aadhaar update demand in the next 14 days to transition to proactive service management. Short-term forecasts and early-warning signals were generated using the trends in Aadhaar Load Score in the past.



The 14-Day Forecast visualization indicates the way the predicted Aadhaar load changes with time. Values that are high in the first forecast window imply that there is an imminent pressure in services whereas decreasing

values imply normalization. This prediction offers lead time that is actionable and therefore the administrators can prepare resources before the queues build or the quality-of-service drops, and not after the congestion has built. Whenever the forecasted load exceeds predetermined stress limits, early-warning flags are created, and PIN codes are identified, which demand preventive measures.

Optimization and Decision Mapping (Phase 4)

Time-series forecasting models were used at the PIN-code level to forecast the Aadhaar update demand in the next 14 days to transition to proactive service management. Short-term forecasts and early-warning signals were generated using the trends in Aadhaar Load Score in the past. The 14-Day Forecast visualization indicates the way the predicted Aadhaar load changes with time. Values that are high in the first forecast window imply that there is an imminent pressure in services whereas decreasing values imply normalization.

★ PIN 431745 shows upcoming stress. Recommended action: Deploy mobile Aadhaar van.

This prediction offers lead time that is actionable and therefore the administrators are able to prepare resources before the queues build or the quality-of-service drops, and not after the congestion has built. Whenever the forecasted load exceeds predetermined stress limits, early-warning flags are created, and PIN codes are identified, which demand preventive measures.

Visualisation and Dashboard Development (Phase 5)

To make the analytical insights to be translated into the practical administrative value, all the outputs of the preceding stages were incorporated into an interactive and multi-level dashboard that should allow clarity, interpretability, and usability in the real world by the officials of the UIDAI.



The dashboard facilitates micro-level decision-making and macro-level decision-making using two supporting views.

PIN-wise Analytical View

On the PIN-code level the dashboard displays:

Aadhaar Load Trend plots of the service pressure change with time.

Structural anomalies, spikes in overload and silent services failures are highlighted in Anomaly Timelines

14-day Demand Forecasts that have early indicators of future stress.



Intervention Impact comparisons, which indicate the load, as predicted before and after suggested actions.

The perspective allows administrators to know why a certain PIN is under threat, how demand can vary and how the suggested interventions can alleviate service pressure.

View of National Priority Operations.

On a national level, the dashboard brings together the insights to aid in the strategic planning:

A national priority map of the best [?]80 high-impact codes of PINs that need to be intervened in.

Colour-based indicators of the suggested measures like the deployment of mobile Aadhaar vans or lengthening the working hours of the centre.

Key performance indicators (KPIs) describing the number of priority PINs, states served and type of interventions needed.

A sortable and filterable priority table with the ability of officials to drill down to state, risk level, or action type.

Distribution charts on a state-by-state basis, which are fair and have a balanced geographical coverage.

The export priority lists are availed by the download options to assist in operational planning and coordination.